CHAPTER 2

Ancient Greek Science and Psychology

THE ORIGINS OF PSYCHOLOGICAL KNOWLEDGE MAY BE SAID TO BE AS old as humankind. From as early as recorded time, men and women have speculated about the nature and source of psychological states and processes and their relationship to human behavior. Theoretical reflections on sensation, memory, and dreaming, for example, are to be found in many ancient works, such as the Hindu sacred texts known as the Vedas (which precede the first millennium BCE) and the Assyrian “dream books” (from around the fifth millennium BCE).

Many early cultures, such as the Egyptian and Babylonian, tried to understand human psychology and behavior in terms of the activity of some immaterial “spirit” or “soul,” usually intimately associated with breath and with the action of the heart and lungs. The Greek term *psyche,* from which the term *psychology* is derived, is etymologically tied to words signifying breath (*pneuma*) or wind (Onians, 1958). There is nothing especially remarkable about this. At a basic level of observation, it is obvious that whatever enables the human organism to act in a purposive fashion is intimately associated with the action of the heart and lungs. When activity in these organs ceases, so also does the activity of the human organism.

Many early theories that postulated immaterial spirits or souls also maintained that such entities could enjoy a life after death in some spiritual realm. However, not all early theories were committed to the notion of an afterlife, and for those that were, it was often an impoverished and literally shady sort of thing. In Greek mythology, for example, the dead survived as shadows of their former selves, which could only be temporarily revived via blood sacrifice.

Beliefs in immaterial spirits or souls are often characterized as **animistic** and are to be found in many so-called primitive cultures today. However, we should guard against the rather condescending assumption that all earlier cultures explained mind and behavior in terms of immaterial spirits or souls and that humans came to a proper understanding of these matters only with the development of psychological science, since such assumptions can seriously prejudice our approach to the history of psychology. Although many ancient thinkers did embrace theories about immaterial spirits or souls, their psychological understanding was far more sophisticated—and materialistic—than is usually acknowledged. Indeed, as will be argued in this chapter, Aristotle's conception of mental states and processes as the functional capacities of complex material bodies approximates our contemporary cognitive psychological conception in critical respects.

This is not to presume the superiority of our contemporary cognitive psychology, which is a matter of lively contention, still less to maintain that modern conceptions of the psychological developed in tandem with historical advances in scientific methodology. On the contrary, as will be noted in later chapters, Aristotle's sophisticated conception of the psychological was one of the casualties of the scientific revolution in Europe during the 16th and 17th centuries.

GREEK SCIENCE

Although beliefs about mind and behavior are as old as humankind, systematic accounts only began to emerge with the development of theoretical science in ancient Greece. The origins of Greek science can be traced to earlier developments in Babylon, Egypt, and Phoenicia, where arithmetic, geometry, and astronomy flourished, and in India and China, where astronomy was well advanced by the second millennium BCE. Some time around the seventh century BCE, the Greeks exploited these developments to forge a mental set that we recognize as a precursor to scientific thinking. This involved a new level of abstract, critical, and speculative thought that within three centuries transformed the intellectual environment of the ancient world. The reasons why such protoscientific thought emerged in ancient Greece are obscure. Increased literacy, facilitated by the appropriation of the Phoenician alphabet, no doubt played a role, as did the unusually liberal (for the time) political structure of the federation of city-states that formed ancient Greece. However, such features seem insufficient in themselves to explain the intellectual revolution that the ancient Greeks produced. As Bertrand Russell once remarked, "nothing is so surprising or difficult to account for as the sudden rise of civilization in Greece" (1945, p. 3).

A distinctive feature of many early Greek thinkers was their rejection of supernatural or religious forms of explanation and their advocacy of naturalistic and mathematical forms of explanation. Yet early Greek "science" was largely speculative. It was loosely based upon empirical evidence and rarely based upon experimentation in the modern sense of manipulative intervention and control. One reason for this, which appears to have extended into the medieval period, is that the forms of intervention required for empirical studies in physics, chemistry, and biology were dismissed as "mechanical" or "servile" arts, suitable only for slaves, as opposed to "liberal arts" such as logic and rhetoric, the approved pursuits of freemen. The Greek historian Xenophon (c. 431–c. 355 BCE) epitomized this attitude: "The mechanical arts carry a social stigma, and are rightly dishonored in our cities."

Nonetheless, early Greek science was often directed to the explanation of puzzling empirical effects and occasionally did employ simple forms of manipulative experimentation, especially in medicine (Lloyd, 1964). The early Greeks advanced very general theories about the nature of reality that appeared to accommodate their sensory experience of the world. If their practice sometimes appears questionable, it is well to remember that they were just starting out. There was already a wealth of empirical effects to be explained, and experimentation was premature given the tentative nature of their theories.

Early Greek science was also critical only in a limited sense. Early Greek theorists offered their theories as speculative hypotheses and expected other theorists to offer alternative hypotheses. Although they offered arguments and analogies in support of their speculative hypotheses, only a few criticized the arguments of their opponents. The systematic criticism of arguments was a later Greek development pioneered by Socrates, Plato, and Aristotle, and the rigorous empirical testing of competing theories was a much later historical development.

However, in early Greek science we find the development of two broad theoretical perspectives that are important components of scientific thought. **Naturalism** is the view that the universe is best explained in terms of material elements and processes. **Formalism** is the view that the universe is best explained in term of formal or mathematical relations. These two perspectives, in conjunction with the later emphasis on the empirical and experimental evaluation of theories, constitute our modern conception of science.

While one should not exaggerate the degree to which these early Greek thinkers anticipated modern science, many of the conceptual features of their theoretical systems are common to modern scientific theories: for example, the exploratory and explanatory use of analogies and the assumption that all apparent change and development is ultimately the alteration of fundamental enduring elements, be they material particles or forms of energy. These early Greek thinkers also extended their theoretical systems to offer rudimentary explanations of biological development and psychological functioning.

Although the Greek term *psyche* is generally translated as "soul," the reader should guard against associations with the contemporary English term. The Greeks conceived of the **psyche** as the general principle of life in animate beings, which included but was not restricted to psychological capacities. The Greek psyche cannot be presumed to be immaterial in nature, like the enduring entities of religions committed to a spiritual afterlife. Although some Greek theorists did maintain that the psyche is immaterial and capable of surviving the destruction of the material body, others denied that this was the case.

THE NATURALISTS

Many early Greek thinkers tried to explain the workings of the universe in terms of its material elements and processes, as opposed to supernatural or religious explanations in terms of immaterial spirits or gods. They represented the beginning of the naturalistic or materialist tradition in science, which attempts to identify the fundamental elements of the natural world. The Greeks characterized the fundamental element as **physis**, and persons who developed systematic theories about the fundamental element (or elements) came to be known as *physicists*.

Thales

Thales (c. 624–c. 546 BCE) of Miletus is generally considered to be the first major theorist in this tradition. He was the founder of what has come to be known as the **Ionian school**, because its principal advocates came from the Ionian federation of city-states, located in what is now the southwest coast of Turkey. Thales declared that the fundamental element is water. This was not an unreasonable speculation, since water can manifest itself as a liquid, solid (frozen), or gas (evaporated) and is essential to all forms of life. It seemed a plausible enough candidate for the basic element that composed all other complex entities.

Thales gained a reputation, common to many abstract thinkers, as a man with his head in the clouds. Aristotle recounts the story of how Thales walked into a well because he was so preoccupied with his study of the stars (Kirk, Raven, & Schofield, 1983, p. 80). Yet he was no ivory tower theorist. Using astronomical calculations to anticipate a record olive crop, he cornered the market in olive presses and made a small fortune leasing them out. He served as an army engineer and was famous in antiquity for having predicted an eclipse of the sun during a battle between the Medes and Lydians. He also is credited with having introduced geometry to ancient Greece, although much of his grounding in the subject was likely derived from his travels in Egypt and Babylon.

The distinguishing feature of Thales' thought was his introduction of abstract, critical, and speculative modes of theorizing. The theses he advanced were offered as hypotheses, not as religiously grounded dogmas. In this respect he may be said to have initiated the critical tradition of scientific thinking. Other Ionian theorists felt free to reject his speculations and to offer their own in critical competition.

Anaximenes

Anaximenes (c. 588–c. 524 BCE) of Miletus postulated that the fundamental element is air, possibly because he was impressed by the infinite malleability of air and the phenomenon of condensation. He developed his theory to explain some puzzling empirical effects, such as the fact that we blow slowly on our hands with an open mouth to warm them, but blow quickly on hot drinks with pursed lips to cool them (Barnes, 1979a, p. 49). Anaximenes accounted for these effects by claiming that properties such as temperature are a function of the density of the constitutive air: The rarefied air from our open mouths is warm, whereas the condensed air from our compressed lips is cooler.

Anaximenes was one of the first to offer explanations of the nature and properties of physical particulars in terms of modifications of an underlying primary element. He explained the nature and properties of clouds, rocks, and human bodies, for example, in terms of their composition by air of different densities, through the processes of rarefication and condensation. According to Anaximenes, rarefied air becomes fire; progressively condensed air becomes wind, then cloud, then water, stone, and so forth. He claimed that the earth itself was formed by the condensation of a vast mass of air, which is ever present in unlimited quantity, and that especially rarefied air constitutes the psyche of living beings. His rudimentary attempt to explicate differences in *qualitative* properties such as temperature and color in terms of *quantitative* properties such as density presaged the distinctively quantitative foundations of modern physical science.

Heraclitus

Heraclitus of Ephesus (c. 540–c. 480 BCE) declared that the fundamental element is fire (or firelike) and maintained that all physical particulars are modifications and alterations of the fundamental and enduring fiery element:

> This world neither any god nor man made; but it always was and is and will be, an ever-living fire, kindling in measures and being extinguished in measures.
>
> —(Barnes, 1979a, p. 61)

For Heraclitus, this included other "elements" such as water, air, and earth, as well as more complex physical particulars such as rocks, trees, animals, and planets. He maintained that condensed fire becomes moist and forms water; solidified water turns to earth; and so forth. The psyche of a human being is composed of fire: It comes from water and returns to water at death, except for a few particularly virtuous souls (such as soldiers slain in battle) who join the cosmic fire. Heraclitus characterized the dry psyche as healthy and the wet psyche as unhealthy: The drunken man behaves like a foolish boy because his psyche is moist.

Heraclitus was primarily concerned to explain the phenomenon of change. He maintained that the natural world is in constant flux, like the waters of a river:

> On those who step into the same rivers, different and different waters flow.
>
> —(Barnes, 1979a, p. 66)

We ordinarily distinguish between enduring physical particulars, such as stones and trees, and transient entities, such as rivers and clouds. However, Heraclitus maintained that the apparent stability and continuity of physical particulars is illusory, because everything is constantly changing:

> And some say not that some existing things are moving, and not others, but that all things are in motion all the time, but that this escapes our perception.
>
> —(Kirk, Raven, & Schofield, 1983, p. 195)

He claimed that the engine of flux is the constant strife or "war" between polar opposites such as hot and cold, wet and dry, and light and dark.

Heraclitus's fundamental vision of the natural world is now commonplace in scientific thought. We recognize that multiplicity and change underlie apparent unity and continuity: We believe that the solid oak chair is really constituted by a multiplicity of atoms (or atomistic wave-packets) and that the cells of our enduring bodies are continuously being replaced.

Empedocles

Empedocles (c. 495–c. 435 BCE) of Acragas denied that any of the four observable physical elements are more fundamental than any other and developed his theory of the "four elements" (Figure 2.1). He held that fire, air, earth, and water are the eternal and irreducible "roots" that constitute all physical particulars: Combined in one proportion they constitute bone, in another proportion they constitute blood, and so forth (Kirk, Raven, & Schofield, 1983, p. 302). He maintained that the processes of combination and dissolution of these elements are governed by the cosmic principles of love and strife, in a continuous cycle of change and development.

Like Heraclitus, Empedocles held that these basic elements and forces are eternal. They account for the creation, temporary endurance, and destruction of all physical

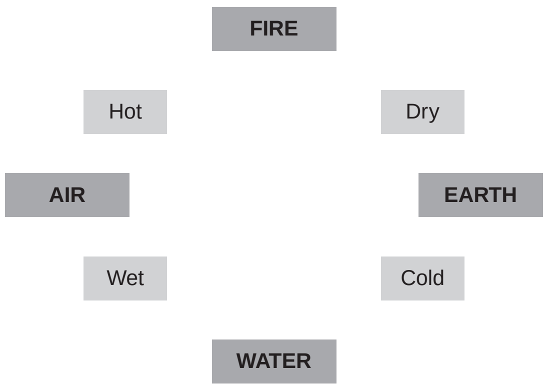


FIGURE 2.1 Empedocles: the four elements.

particulars, such as planets, oceans, and animals, which are merely the successive combination and separation of these basic elements in different proportions:

> Double is the birth of mortal things and double their failing; for the one is brought to birth and destroyed by the coming together of all things, the other is nurtured and flies apart as they grow apart again. And these things never cease their continual interchange, now through love all coming together into one, now again each carried apart by the hatred of strife.
>
> —(Kirk, Raven, & Schofield, 1983, p. 287)

Like Heraclitus and Anaximenes, Empedocles maintained that all apparent creation and destruction in the natural world is merely the alteration of the fundamental enduring elements of the material substratum: "insofar as they never cease their continual interchange, thus far they exist always changeless in the cycle" (Kirk, Raven, & Schofield, 1983, p. 287).

His theory was enormously influential, especially in medicine and psychology. Empedocles maintained that health consists of the proper balance of the four elements in our physical bodies and blood. The Greek and Roman physicians Hippocrates and Galen developed this account into the theory of physiological and psychological "humors," and similar principles of equilibrium or homeostasis are to be found in a great many later biological and psychological theories.

Empedocles held that blood contains the four elements in almost perfect combination and consequently identified blood as the medium of perception and thought. According to Empedocles, physical bodies emit effluences, or **eidola**, in the form of faint copies of themselves. Perception occurs when eidola enter the blood through the pores of the skin and their compositional elements match up with like elements in the blood (fiery elements of the eidola match up with fiery elements in the blood, and so forth), generating images in the heart:

> *Nourished in a sea of churning blood*
> *where what men call thought is especially found—*
> *for the blood about the heart is thought for men.*
>
> —(Barnes, 1987, p. 191)

In his account of the origin of animals and humans, Empedocles developed a rudimentary theory of evolution. He described how the parts of animals (constituted by the elements in different proportions) were originally combined in a random fashion, resulting in a variety of hybrid forms:

> But as one divine element mingled further with another, these things fell together as each chanced to meet other . . . , and many other things besides these were constantly resulting.

> Many creatures were born with faces and breasts on both sides, man-faced ox-progeny, while others sprang forth as ox-headed offspring of man, creatures compounded partly of male, partly of the nature of female, and fitted with shadowy parts.
>
> —(Kirk, Raven, & Schofield, 1983, p. 304)

Empedocles' theory anticipated some of the distinctive features of Darwin's later theory of evolution, such as the random mutation of biological forms described above (albeit fantastically). More significantly, as Aristotle clearly recognized, Empedocles provided an account of how "suitably formed" combinations that developed *by chance* would be naturally selected over time and produce adapted species that appeared, but *only* appeared, to have been purposely created:

> Wherever, then, everything turned out as it would have if it were happening for a purpose, there the creatures survived, being accidentally compounded in a suitable way; but where this did not happen, the creatures perished and are perishing still, as Empedocles says of his "man-faced ox-progeny."
>
> —(Kirk, Raven, & Schofield, 1983, p. 304)

Empedocles was acclaimed as a great thinker during his lifetime, attaining almost godlike status. He believed that the psyche was composed of all four elements, which could be recombined to constitute a different psyche in different generations (the principle of metempsychosis). Thus Empedocles believed that he had been different beings in his former lives:

> *For already have I once been a boy and a girl*
> *And a bush and a bird and a silent fish in the sea.*
>
> —(Barnes, 1987, p. 196)

The Atomists: Leucippus and Democritus

The theories of the Greek atomists Leucippus (c. 500–c. 450 BCE) and his disciple Democritus (c. 460–c. 370 BCE) represent the culmination of the naturalist tradition in early Greek thought. They maintained that the ultimate constituents or elements are "atoms and the void." Of Leucippus little is known, and most of what we know of Greek atomism is based upon the views attributed to his pupil Democritus. Democritus was called the "laughing philosopher," supposedly because he was so amused by human folly. According to a contested legend, he blinded himself in order to secure his happiness, in the belief that his blindness would eliminate his desire for women.

The Greek atomists maintained that atoms are solid particles that differ only in their properties: "some of them are scalene, some hooked, some hollow, some convex" (Barnes, 1979b, p. 41). They bind together on contact, through "overlappings and interlockings of the bodies" (Barnes, 1979b, p. 41). They are infinite, indivisible, and invisible, and their movements and bindings in the void are responsible for the creation and destruction of physical particulars:

> They move in the void . . . and when they come together they cause coming to be, and when they separate they cause perishing.
>
> —(Kirk, Raven, & Schofield, 1983, p. 407)

They held that the combination of atoms in empty space explains the properties of physical particulars. For example, they explained the different weights of different types of physical particulars in terms of their different densities and claimed that the celestial bodies were formed by the ignition of dense masses of atoms compressed into rapidly moving vortexes.

Democritus developed a theory of perception similar to Empedocles', based upon the atomic hypothesis. He claimed that thin films of atoms shaped in the form of physical bodies emanate from their surface. Our sense organs receive these eidola and interact with (highly mobile) fire atoms in our brain, which form copies of physical bodies. We perceive physical bodies and their properties when similar arrangements of atoms form in our sense organs and brain. Democritus maintained that the psyche is itself composed of fine fiery atoms, which are dispersed with the dissolution of the living body.

Like Empedocles and Heraclitus, the Greek atomists claimed that all perceived creation and destruction in the natural world is merely the alteration of the fundamental and enduring elements of the material substratum and that complexity and change underlie the apparent unity and stability of physical particulars. However, they went one step further, anticipating a distinctive feature of modern science.

Most Greek naturalists accepted the perceived properties of physical particulars at face value and attempted to explain them in terms of the arrangement and alteration of their fundamental material components. They maintained that our senses do not reveal the fundamental nature of reality, but they did not generally deny the reality of the empirical appearances they tried to explain. However, the Greek atomists claimed that many of the perceived properties of physical particulars are not genuine properties but are only their effects on our sense organs and nervous systems.

They made a distinction between those properties that physical particulars have independently of our perception of them, such as size, shape, and motion, and those "properties" that are merely the effects they produce in the sense organs and nervous systems of sentient beings, such as color, taste, and smell. Consequently, they maintained that while there are atoms with shapes, sizes, and motions in the void, there are strictly speaking no colors, smells, tastes, sounds, or textures. Democritus famously maintained that

> By convention sweet and by convention bitter, by convention hot, by convention cold, by convention color: in reality atoms and void.
>
> —(Barnes, 1987, p. 253)

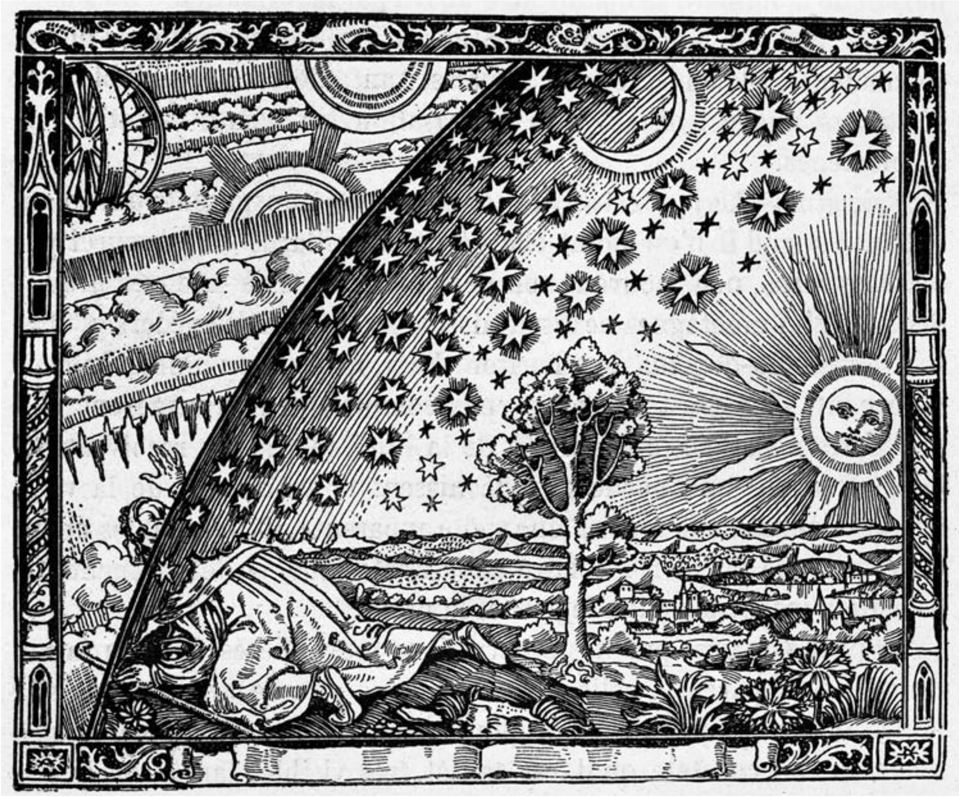
This distinction between what later came to be known as **primary qualities** and **secondary qualities** was made by most of the pioneers of the scientific revolution in Europe in the 16th and 17th centuries. Like these later scientists, the Greek atomists explained the generation of secondary qualities such as taste as a causal product of the primary qualities of atoms, such as their shape and size:

> Sour taste comes from shapes that are large and multi-angular and have very little roundness; for these, when they enter the body, clog and bind the veins and prevent their flowing. . . . Bitter taste comes from small, smooth, rounded shapes whose periphery does have joints; that is why it is viscous and adhesive.
>
> —(Barnes, 1979b, p. 71)

The Greek atomists also maintained that the universe is governed by rigidly deterministic laws. Whatever happens in the natural world is determined by natural necessity, as an inevitable consequence of the arrangements and motions of the constituent atoms. There is no room for randomness or choice in the purely mechanistic universe of the Greek atomists: There are only atoms and the void. Still, Democritus was no fatalist. He maintained that happiness is the goal of life, which is best attained through self-control and moderation, including the general avoidance of the pleasures of the flesh: “For men gain contentment from moderation in joy and a measured life” (Barnes, 1987, p. 269). Epicurus (341–271 BCE) later developed this sophisticated form of hedonism, although the position now popularly known as epicureanism bears little relation to the doctrines of either, being commonly associated with the maximal satisfaction of sensual desires.

THE FORMALISTS

Although the early Greek naturalists generally accepted the evidence of sense experience, they gradually came to conceive of theoretical knowledge as the discernment of the underlying reality beyond or “behind” sensory appearances. A similar notion seems to have motivated many of the theorists of the formalist schools developed by Parmenides (b. c. 515 BCE) and Pythagoras (b. c. 570 BCE), who were more deeply skeptical of our sense experience of the world. According to these formalist theorists, the changing world of sense experience is illusory, and



The natural philosopher discovers the reality beyond sensory appearances. *The Heavens.* Camille Flammarion. From *L'atmosphére metorologie populaire,* 1898, in the style of the 1500s, woodcut.

the changeless reality beyond or "behind" sensory appearances can be grasped only by reasoning and logic. Formalist theories were based mainly upon logical deduction and argument, in contrast to the empirical puzzles and theoretical analogies exploited by naturalists.

Parmenides

Many formalists denied the reality of change. According to Parmenides of Elea, reality is unitary, unchanging, motionless, indivisible, and eternal. In his extended poem *On Nature*, he claimed that we can deduce that reality is an eternal unity:

> *. . . that being, it is ungenerated and undestroyed,*
> *whole, of one kind and motionless, and balanced.*

> *Nor was it ever, nor will it be; since now it is, all together, one, continuous.*
>
> —(Barnes, 1979a, p. 178)

Parmenides contrasted the "way of truth" with the "way of opinion," the way reality appears to our sense experience, as a created multiplicity of moving, changing objects.

Parmenides was acutely conscious of the conflict between reason and sense experience and maintained that sense experience is illusory. He claimed that the perfect, eternal, and unchanging intelligible world, unlike the imperfect, temporary, and changing sensible world, can be known only through the exercise of reason, which gets us to the reality beyond or "behind" sensory appearances. Parmenides is often credited with the development of the **dialectic method**: the systematic exploration of arguments for and against opposing positions. He was the primary theorist of what came to be known as the **Eleatic school**, so-called because its members came from Elea, a Greek settlement in southern Italy. Other members of the school included Zeno of Elea (b. c. 490 BCE) and Xenophanes (b. c. 570 BCE), who famously ridiculed the petty foibles and frailties of the Greek gods.

Since Parmenides maintained that reality is unchanging, he is often characterized as a theorist of *being*, in contrast to Heraclitus, who maintained that reality is constantly changing and who is often characterized as a theorist of *becoming*; the contrast between their theories is often referred to as the debate between **being vs. becoming**.

Zeno of Elea

Zeno of Elea developed a famous series of arguments in support of the Parmenidian position. These arguments were designed to demonstrate the illusory nature of sense experience and commonsense assumptions about multiplicity and change based upon it. According to Plato, Zeno tried to protect the Parmenidian "affirmation of the one" from critical ridicule by demonstrating that the "hypothesis of the existence of many, if carried out, appears to be still more ridiculous than the hypothesis of the existence of one" (Barnes, 1979a, p. 233).

Zeno advanced a number of **reductio ad absurdum** arguments, which purport to demonstrate the falsity of commonsense assumptions about multiplicity, change, and motion by demonstrating that they lead to false or absurd consequences. The most famous of these, commonly known as Zeno's paradoxes, purport to demonstrate the illusory nature of motion. The best known is the tale of the race between Achilles and the tortoise, who is given a head start because he cannot move as fast as Achilles. Whenever Achilles reaches the tortoise's starting point, the tortoise has moved on to another point; whenever Achilles reaches that point, the tortoise has moved on to another; so Achilles can never catch up with the tortoise. As Aristotle put it:

> This says that the slow will never be caught in running by the fastest. For the pursuer must first get to where the pursued started from, so that it is necessary that the slower should always be some distance ahead.
>
> —(Barnes, 1979a, p. 273)

Pythagoras

Parmenides, Zeno, and other members of the Eleatic school maintained that theoretical understanding could be attained only through abstract reasoning. Pythagoras (c. 572–497 BCE) and his followers agreed, but went one step further and maintained that ultimate reality *is itself abstract in nature*, being constituted by mathematical harmonies and ratios. They maintained that the illusory world of sense experience is merely the manifestation of fundamental mathematical harmonies and ratios. In contrast to the naturalists, who sought theoretical understanding of the basic material constituents of the universe, the Pythagoreans sought understanding of its basic formal principles.

Pythagoras formulated ontological and epistemological distinctions between the abstract objects of mathematics and logic and the concrete physical particulars of the natural world that are the objects of sense experience. He claimed that mathematical and logical relations are perfect, eternal relations that exist independently of physical particulars and that knowledge of the abstract truths of mathematics and logic can be attained directly through the exercise of pure reason, independently of sense experience. For example, it is eternally true that the angles of a triangle add up to 180°, whether or not any physical triangles have ever existed or been perceived by sentient beings. Moreover, this truth can be known (at least in principle) by a rational being without any sense experience of physical triangles. In contrast, the objects of sense experience, which are subject to creation, destruction, and change, are imperfect and cannot be truly known. Pythagoras's distinction between the "intelligible" and the "sensible" worlds exerted a powerful influence on many later thinkers, including Plato.

Pythagoras was a **dualist**, who maintained that mind and body are distinct entities. He distinguished between the immortal psyche, which can rationally apprehend the intelligible world, and the corruptible material body in which it is temporarily imprisoned. This conception of the relation between the psyche and the material body, which likely had its origins in the mystery religions of Greece and Egypt, influenced many ancient and medieval theorists, such as Plato, Plotinus, Saint Augustine, and Avicenna.

Pythagoras and his followers made major contributions to mathematics, astronomy, and physics. They formulated the basic principles of arithmetic and geometry later described in Euclid's *Elements*. Pythagoras explained the formation of the universe in mathematical terms, as an imposition of limit on the limitless, and one of his followers Philolaus is reputed to have been one of the first to conceive of Earth as a moving planet. Pythagoras is also credited with the discovery of the musical scale, by developing a set of mathematical laws describing the harmonic ratios of vibrating strings of different lengths. This latter achievement seems to have inspired his view that everything in the universe can be explained in terms of the principles of mathematical harmony.

Pythagoras was a charismatic leader. The school he founded, while dedicated to the study of mathematical harmonies and ratios, was as much a moral as an intellectual brotherhood. His rational depreciation of the sensible world and sense experience was matched by a moral depreciation of the material body and sensual pleasure. Pythagoras and his followers believed in the immortality and transmigration of the psyche. According to Pythagoras, each human psyche is possessed of its own divinity and goes through cycles of rebirth in vegetable, animal, and human form, which it can remember. Release from the bodily prison and cycle of rebirth can be attained only by the purification of the psyche through rational contemplation of the intelligible world, by which it may attain eventual union with the "world soul."

Pythagoras and his followers repudiated the sensual pleasures of the material body, and committed themselves to an ethic of abstinence and self-discipline. This included the prohibition of meat or beans, since both might include transmigrated souls, and adoption of various measures designed to liberate the psyche from its bodily prison. Their conception of the body as a temporary prison subject to physical and moral decay through age and sensual corruption exerted a powerful influence in the following centuries, most notably on early Christian theology.

Pythagoras and his followers took their commitment to mathematical harmony to mystical extremes. They associated justice with the number four and reason with the number one; and they reputedly drowned Hippasus of Metapontum, a fellow Pythagorean, because of his discovery of irrational numbers (Blackburn, 1996, p. 173). Nonetheless, the Pythagorean school marks the beginning of the mathematical tradition that inspired Galileo to characterize mathematics as the language of science, Kepler to try to compose the "harmony of the celestial spheres" based upon the geometry of the regular solids, and Newton to spend much of his life working on the numerology of the biblical Book of Daniel.

Although Pythagoras treated the psyche as an entity capable of surviving bodily death, he identified the brain as the bodily organ of thought. He also developed an account of physical and psychological health based upon the harmonious blending of bodily elements. Since Pythagoras maintained that physical and psychological disorders derived from the disruption of bodily harmony, his recommended treatments were designed to restore harmony and included regulated diet, exercise, and music. Empedocles' treatment of health as harmony between the "four elements" was likely based upon the Pythagorean account.

THE PHYSICIANS

Early Greek medicine was based upon religious mysteries and practiced by temple priests who kept the secrets of Asclepius, the Greek god of medicine. Treatments consisted largely of sleep, suggestion, diet, and exercise. These forms of **temple medicine** were challenged by the schools of Alcmaeon and Hippocrates, who rejected religious beliefs and mystical practices in favor of naturalistic theories and treatments based upon observation (however rudimentary).

Alcmaeon

Alcmaeon (fl. c. 500 BCE) founded a school of medicine in Croton, in southern Italy, but rejected the theory and practice of temple medicine. He established that the brain is the center of perception and cognition: He dissected the human eye and brain and traced the optic nerves from the retina to the brain (Lloyd, 1991). Alcmaeon also developed the influential theory of **animal spirits**, conceived of as the material carriers of nerve impulses.

Alcmaeon rejected mystical and religious medical accounts and advanced a theory of health and disease based upon the properties associated with the four elements of Empedocles, such as hot and cold, dry and moist, and so forth. He claimed that health consists of the proper balance of these properties and that disease is caused by imbalance. Since an excess of heat causes fever, it should be treated by cooling the patient; since an excess of dryness causes dehydration, it should be treated by increasing the intake of liquids. Alcmaeon believed that extremes of the elemental properties are the cause of death.

Hippocrates

Hippocrates (c. 460–377 BCE), who is often called the father of medicine, was born on the Greek island of Cos, off the coast of modern Turkey. He studied at the great center of temple medicine there, but founded his own medical school when he later came to reject temple medicine and attained a celebrated reputation as a healer and teacher. Only a few of the medical writings attributed to Hippocrates, the *Corpus Hippocraticum*, were probably written by him, so it is hard to distinguish his individual contribution from those of his followers. Accordingly, it is best to treat him as the representative leader and spokesman of the medical school of Cos.

Like Alcmaeon, Hippocrates claimed that the brain is the center of psychological capacities:

> It ought to be generally known that the source of our pleasure, merriment, laughter and amusement, as of our grief, pain, anxiety and tears is none other than the brain . . . it is the brain too which is the seat of madness and delirium.
>
> —(Lloyd, 1983, p. 248)

He maintained that epilepsy, the so-called sacred disease, is a disorder of brain function and that popular explanations in terms of divine possession are pseudo-explanations grounded in ignorance:

> I do not believe that the "Sacred Disease" is any more divine or sacred than any other disease, but, on the contrary, has specific characteristics and a definite cause. . . . It is my opinion that those who first called this disease "sacred" were the sort of people we now call witch-doctors, faith-healers, quacks and charlatans. . . . By invoking a divine element they were able to screen their own failure to give suitable treatment and so called this a "sacred" malady to conceal their ignorance of its nature.
>
> —(Lloyd, 1983, pp. 237–238)

Hippocrates developed Alcmaeon's conception of health as balance into the influential theory of the **four humors**—yellow bile, blood, black bile, and phlegm—supposedly formed from the four elements postulated by Empedocles (Figure 2.2). According to this theory, yellow bile is formed from fire, blood from air, black bile from earth, and phlegm from water. According to Hippocrates, health derives from the proper balance of these humors, and disease from their imbalance:

> Health is primarily that state in which these constituent substances are in the correct proportion to each other, both in strength and quantity, and are well mixed. Pain

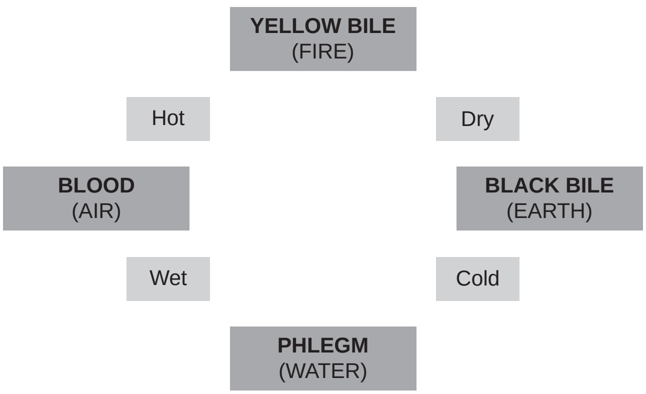


FIGURE 2.2 Hippocrates: the four humors.

> occurs when one of the substances presents either a deficiency or an excess, or is separated in the body and not mixed with the others.
>
> —(Lloyd, 1983, p. 262)

Consequently Hippocrates recommended treatments designed to restore humoral balance, such as appropriate diet, rest, exercise, bathing, massage, and laughter.

Hippocrates is often characterized as an early practitioner of **holistic medicine**, because he emphasized the natural healing power of the body and the need to treat physical and psychological disorders as disorders of the whole body. However, he was not averse to medical intervention. He recommended trepanning for the relief of brain tumors and bloodletting to treat disorders supposedly caused by an excess of blood, a popular medical practice that continued until the 18th century. Hippocrates also recognized that physical and psychological diseases and disorders could sometimes be relieved through mere faith in the competence and commitment of the physician (see Frank, 1973, for a modern account):

> For some patients though conscious that their condition is perilous, recover their health simply through their contentment with the goodness of the physician.
>
> —(Jones, 1923, p. 319)

Hippocrates' accounts of physical and psychological disease and disorder are remarkable in their diagnostic detail. In the *Art of Healing,* he provided extensive descriptions of arthritis, epilepsy, mumps, and tuberculosis and of paranoia, phobia, depression, mania, and hysteria. On the basis of his studies of brain damage and paralysis, he established the contralateral control of the body by the hemispheres of the brain.

Hippocrates and his followers developed a code of ethics for physicians, now known as the Hippocratic oath, which contemporary physicians vow to follow. This includes the injunction not to refuse treatment to patients who cannot afford to pay for it and to decline payment for the treatment of patients in financial straits.

THE PHILOSOPHERS

As Greece and Athens entered their "golden age," many thinkers came to reflect on the epistemological foundations of naturalism and formalism. In an intellectual arena in which competing theories were supported by arguments, skill in argumentation came to be appreciated and valued for its own sake.

The **Sophists** were professional teachers of rhetoric and logic, who charged their students for instruction in the art of persuasion. They were skeptical about the possibility of human knowledge, although they also professed it for a fee. The term *Sophist* was originally applied to any wise man, but it acquired negative connotations when Plato caricatured some Sophists as money-grabbing charlatans. Yet they offered practical advice as well as instruction in argument. For example, Antiphon is reputed to have offered an early form of "verbal" psychotherapy based upon interactive dialogue to those suffering from grief and melancholy (Pivnicki, 1969; Walker, 1991).

They also raised troubling questions about claims to knowledge of the natural world based upon sense experience. Protagoras of Abdera (c. 490–c. 420 BCE) based his claim that "Of all things man is the measure" (Barnes, 1979b, p. 239) upon the variability of sense experience: What seems cold or loud to one person may seem warm or quiet to another person, or to the same person at a later time. His intent may have been only to affirm the authority of immediate sense experience, but he is commonly taken to have advocated a form of **relativism**, according to which what is true is relative to what any individual perceives or judges to be the case. Gorgias of Leontini (c. 485–380 BCE) argued that we could not have knowledge of anything beyond or "behind" sense experience. It was against such forms of relativism and skepticism that Socrates and Plato advanced their accounts of objective knowledge.

Socrates

Socrates (c. 469–399 BCE) gave Western philosophy its distinctive focus on the critical analysis of concepts and arguments. His single-minded commitment to the pursuit of wisdom is best exemplified by his famous claim that "the unexamined life is not worth living." Apart from short periods of military service and occasional work as a stonemason, he devoted his life to philosophical disputation with the Sophists, his aristocratic friends, and the people of Athens. Indifferent to fame and fortune, he accepted no fees for his teaching, since he professed not to know anything worthwhile. In 399 BCE he was charged with corrupting the Athenian youth. His consequent trial, imprisonment, and death are movingly described in Plato's *Apology, Crito,* and *Phaedo*.

Socrates focused on ethical questions and taught that virtue is knowledge. He tried to discover the objective essence of courage, justice, knowledge, and virtue through critical examination of proposed definitions in terms of properties common to all their instances. For example, in *Theaetetus,* Plato has him defining knowledge in terms of justified true belief. His method of critical examination has come to be known as the Socratic method.

Plato

Plato (429–347 BCE) was born in Athens of an aristocratic family. His disgust with the politics of his day reputedly led him to conclude that only philosophers are fit to rule, a position that he defended at length in the *Republic*. After the death of his teacher Socrates, Plato traveled extensively. He returned to Athens and founded his Academy in 387 BCE; he enjoyed a successful career as a teacher until his death at the age of 82.

Many of Plato's doctrines were developments of Pythagorean theory. He was a dualist who believed that the psyche is an immortal and immaterial entity temporarily imprisoned in a material body, and he claimed that true knowledge can be attained only when the purified psyche surmounts the corruption of the material body (through self-discipline or death). In the *Republic,* Plato maintained that justice or well-being in the individual and the state is founded upon the harmony of the parts of the hierarchic and tripartite psyche, comprising reason, passion, and appetite. Psychological harmony is achieved when reason controls the passions and appetites; psychological disorder and immorality result when the passions and appetites gain control. In his description of the conflict between reason and the passions and appetites, Plato anticipated the later Freudian contrast between the rational and irrational elements of human personality (Simon, 1972).

Like Pythagoras, Plato distinguished between the intelligible and sensible worlds. He claimed that the abstract forms or ideas elicited via the Socratic method of critical examination are more real than the concrete physical particulars supposedly revealed through sense experience. According to Plato's **theory of Forms**, such abstract ideas are perfect, eternal, and unchanging, in contrast to imperfect, transient, and changing physical particulars. Plato maintained that physical particulars are imperfect copies of the Forms and exist only via derivative "participation" in them. Consequently he distinguished genuine knowledge (*episteme*) derived from rational apprehension of the Forms from mere belief or opinion (*doxa*) based upon sense experience.

In maintaining that knowledge of the Forms can be attained through the exercise of reason independently of sense experience, Plato defended a nativist conception of knowledge based upon the Pythagorean doctrine of the transmigration of the psyche. According to Plato's **theory of recollection**, all knowledge is remembrance of knowledge possessed by the immortal psyche, but temporarily forgotten with each cycle of reincarnation. In a famous passage in the *Meno*, Plato described how Socrates managed to elicit innate knowledge of geometry from an untutored slave boy.

Like Socrates, Plato was grappling with the threats posed to objective knowledge by the relativism and skepticism of the Sophists. Yet his own solution was almost as bad as the original relativist and skeptical threats. The notion that sense experience is illusory and impedes the attainment of genuine knowledge through pure exercise of the intellect cast a dead hand on the development of scientific reasoning in the ensuing centuries, particularly when it was taken up by the early Christian Church fathers.

ARISTOTLE: THE SCIENCE OF THE PSYCHE

Aristotle (384–322 BCE) was born in Macedonia, the son of a royal physician. He entered Plato's Academy at the age of 17 and remained there until Plato's death. After a period of traveling, during which he married and served as tutor to the young Alexander the Great (from 343–340 BCE), he returned to Athens in 335 BCE and founded his own school, the Lyceum. There he instituted research on a wide variety of subjects and created the first great library of antiquity. After the death of Alexander the Great in 323 BCE, Aristotle felt obliged to retire to Chalcis because of the level of anti-Macedonian feeling, lest Athens "sin twice against philosophy" (the first sin being the trial and execution of Socrates). He died a year later.

Aristotle's theoretical contributions range across a wide variety of subjects and cover most of the sciences known in his day (although some parts of his extensive body of work are attributed to his Lyceum students). His theory of the syllogism provided the justificatory basis of logical inference until the late 19th century, when its limitations were finally realized. He was the first Greek theorist to devote a whole work to psychology (*De Anima*), although his psychological contributions range across a variety of works, such as *On Memory*, *On Dreams*, and *Nicomachean Ethics*. He was also the first theorist to reflect critically on the nature of psychological explanation, and the subtlety and sophistication of his account has scarcely been rivaled since.

Yet what was distinctive about Aristotle, and what distinguished him from early naturalists and formalists, was his strong empirical bent. His researches in biology, for example, were based upon a wealth of careful observations of the flora and fauna of the natural world. His detailed descriptions of the embryology of the chick in the *History of Animals*, which were based upon his dissection of eggs and embryos at different stages of incubation, took the subject pretty much as far as it could be taken prior to the development of the microscope.

Aristotle was not afraid to get his hands dirty and was a great collector of biological specimens. According to legend, he spent most of his honeymoon extending his collection of seashells. He got many things wildly wrong, although he usually had some reason for adopting the views he did. We may scoff at Aristotle's belief that the heart rather than the brain is the organ of perception and cognition. Yet it was based upon his observation that the heart is the first organ to manifest activity in the embryological development of the chick and that humans and other animals recover with greater frequency from wounds to the head than wounds to the heart. More important, Aristotle kept an open mind on most theoretical matters. He emphasized that his own theoretical contributions were provisional and based upon the limited development of the sciences of his day and that the last court of appeal for any theory was observation. For example, in discussing the generation (procreation) of bees in the *Generation of Animals* (III, 760b29–33), he qualified his tentative theoretical claims:

> Such appears to be the truth about the generation of bees, judging from theory and from what are believed to be the facts about them; the facts, however, have not yet been sufficiently grasped; if ever they are, then credit must be given rather to observation than to theories, and to theories only if what they affirm agrees with the observed facts.[1]

Theoretical Science

For Aristotle, the goal of theoretical science was the classification of substances and explanation of their properties. He claimed that essential form (*morphe*) constitutes matter (*hule*) as particular substances—a doctrine known as **hylomorphism** (Jager & VanHoorn, 1972). For Aristotle, matter is the basic constituent of all substances, which are individuated as distinct substances—as individual frogs as opposed to individual roses, for example—by their essential form. Thus a rose is matter in the form of a rose; a frog is matter in the form of a frog; a human being is matter in the form of a human being, and so forth.

Aristotle held that the essential form of a substance is only conceptually distinct from its matter. Although the different forms of different types of substances can be identified and distinguished, they do not and cannot exist independently of matter. Plato had maintained that forms exist as autonomous abstract particulars in a realm of ideas, which can be apprehended only through pure reason. In contrast, Aristotle held that forms exist only as the forms of matter. For example, sphericality (the essential property of spheres) can be conceptually distinguished from cubicality (the essential property of cubes), but only exists as the sphericality of materially instantiated substances. Aristotle also claimed that knowledge of forms could be gained only through sense experience, by abstraction from the perceived common properties of classes of substances, such as roses, frogs, and human beings. He consequently denied that we have innate knowledge of common properties or **universals**.

Aristotle also distinguished between potentiality and actuality. He accounted for all change in nature in terms of the process of **entelechy**, by which what is merely potential becomes actual through the realization of its form. This conception of change as a process in which determinate potentiality becomes actualized was best suited to the explanation of biological development, in which what is potentially an oak, an acorn, becomes an oak; what is potentially a chick, an embryo, becomes a chick; and what is potentially an adult human being, a

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[1]All quotations from Aristotle are from Barnes (1995).

FIGURE 2.3 Aristotle: the scala naturae.

neonate, becomes an adult human being. However, Aristotle also extended this account to other natural changes, such as changes in the motion of physical bodies.

Aristotle represented nature as a hierarchically structured order of existents, or **scala naturae**, in which simpler forms of being, or formed matter, serve as the matter for higher forms of being, such as plants and animals (Figure 2.3). At the bottom of the scale is prime matter: It has the potential for form, but in its unformed state is mere potentiality. The simplest types of formed matter or substances, the primary bodies, are the four elements identified by earlier Greek physicists: fire, air, earth, and water. These combine in successively more complex forms to constitute inanimate bodies and living organisms such as plants, animals, and human beings, organized according to genus and species. At the very top of the scale is the **unmoved mover**, or pure actuality, which is responsible for the realization of all things and whose form does not itself require material instantiation. This hierarchical conception of nature as a **great chain of being** (Lovejoy, 1936) exerted a powerful influence on later theorists and served as the foundation of most systems of biological classification until the 18th century.

Aristotle placed human beings at the midpoint of the scala naturae and placed Earth at the center of the universe. According to Aristotle's geocentric (Earth-centered) theory, the sun and other planets traverse concentric orbits around Earth. Aristotle postulated a set of hollow and transparent crystalline spheres that carry the planets in their constant circular orbits. The outermost sphere, the celestial sphere, carries the fixed stars and marks the finite boundary of the universe. Aristotle claimed that celestial objects, unlike objects in the sublunar region that are composed of fire, air, earth, and water, do not undergo substantive change and are composed of a fifth element, which he called the ether.

Aristotle claimed that scientific knowledge is knowledge of first principles, which he held to be necessary truths. He believed that scientific knowledge is originally based upon generalization from observed instances, or **induction by enumeration**; for example, we come to know that "all ruminants with cloven hooves are animals with missing incisor teeth" on the basis of observing instances of ruminants with cloven hooves that have missing incisor teeth. However, he maintained that knowledge of substantial forms is based upon the direct rational intuition of first principles, which enables humans to discriminate the essential from the accidental properties of substances (Losee, 1980). Essential properties of substances are those properties that members of a class must have in order to count as members of that class; accidental properties are properties that members of a class happen to have, but need not have in order to be members of that class. For example, sentience and rationality are essential properties of human beings, whereas differences in their complexion are accidental properties.

Causality and Teleology

Aristotle distinguished four types of causality, which he claimed play a role in the explanation of all existents in the natural world. The **material cause** of an existent is the material in which it is realized. The material cause of a statue of Zeus would be the marble or bronze from which it is created. The **formal cause** of an existent is its essential form, which distinguishes this type of existent from all others. The formal cause of a statue of Zeus would be the structure or shape of the statue, sculpted in the image of Zeus. The **efficient cause** of an existent is the agency responsible for its generation. The efficient cause of a statue of Zeus would be the artist who created it. The **final cause** of an existent is the end or function or purpose for which it exists. The final cause of a statue of Zeus could be the glorification of the gods. Although the preceding example is useful for illustrative purposes, Aristotle maintained that the **four causes** are only distinct in the case of artifacts such as statues. In the case of all other natural existents, the formal cause is also the efficient and final cause.

Modern conceptions of causality tend to equate causality with efficient causality: that is, with conditions sufficient to bring about an effect. Contemporary scientists continue to employ material and formal causal explanations, insofar as they explain the powers and properties of chemical compounds and biological organisms in terms of their composition and structure, but have abandoned final causal explanations in physical and biological science. One of the distinctive features of the scientific revolution in Europe in the 16th and 17th centuries was the rejection of final causal explanations in favor of efficient causal explanations in physics and medicine. Yet Aristotle's science was a comprehensively **teleological science**, in which all natural processes were explained in terms of an ascribed end or goal state (*telos*).

To get the flavor of this, consider how an explanation of photosynthesis in plants might proceed in terms of Aristotle's four causes. We might say that the material cause of photosynthesis is the organic molecules out of which plants are composed. We might say that the formal cause of photosynthesis is the biological structure that distinguishes plants from other types of entities (including the fact that they contain the enzyme chlorophyll), which explains their ability to perform photosynthesis. We might say that the efficient cause of photosynthesis is the action of sunlight on plants, which, given their composition and structure, is sufficient for photosynthesis. And we might say that the final cause of photosynthesis, its end state or function or purpose, is the maintenance of atmospheric oxygen, which sustains those life forms that depend upon oxygen.

Now the material, formal, and efficient causal components of this complex explanation are relatively unproblematic. They map easily onto contemporary forms of explanation in chemistry and biology, which specify enabling and stimulus conditions that are jointly sufficient for an effect or process. However, the last component, the final cause, is alien to modern science, which would treat references to the end or function or purpose of photosynthesis as redundant from the point of view of scientific explanation. Modern biologists would insist that while it is a fortunate consequence for humans and other oxygen-dependent species that photosynthesis contributes to the maintenance of atmospheric oxygen, this is nothing more than a fortuitous effect: It is not the end or function or purpose of anything or anyone.

Similarly, post-Darwinian biologists dismiss explanatory references to the function of the human eye or the long neck of the giraffe. They maintain that the human eye and the giraffe's neck have no function or purpose in themselves, and that their existence can be exhaustively explained in terms of the survival advantages conferred upon organisms possessing them in particular environments. Although modern biologists retain talk of functions as an informal convenience, they hold that such talk can be eliminated from biology without any loss of scientific insight.

Yet Aristotle maintained that all existents have ends or functions or purposes. For Aristotle, this was as true of the motion of physical bodies as it was for biological processes and human behavior. He explained "natural" motion in terms of the "natural resting place" of each of the primary bodies, to which they move when unopposed: Earth moves to the center; and water, air, and fire move to successive spheres about the center (he explained "unnatural motion" in terms of impressed forces). Aristotle maintained that the scala naturae is purposively ordered, with every existent having its fixed place and function, and rejected the evolutionary theory of Empedocles.

Aristotle treated natural processes as **intrinsically teleological**, since he maintained that the ends or purposes are inherent in the processes themselves. Later generations of Christian and Islamic scholars treated them as **extrinsically teleological**, as a product of the ends or purposes of separate beings, such as a supernatural God conceived of as the intelligent designer of the natural world. Yet this formed no part of Aristotle's account. His commitment to teleological science was not a product of his commitment to a belief about intelligent creation, but a generalization of his belief in the inherent directionality of biological development.

Aristotle's Psychology

The basic principles of Aristotle's psychology were developed in *De Anima* ("on the soul"), and are the product of the application of his causal schema to human beings. According to Aristotle, the psyche is the formal cause of a human being, the set of functional properties that constitute certain substances as human beings. The material cause of human beings is the organized organic material out of which they are composed. According to the principle of entelechy, the psyche is the actuality of a material body that potentially has life: It actualizes the potential of the human body to be the living creature that is a human being. As Aristotle put it, the psyche is "the form of a natural body having life potentially within it" (*De Anima*, II, 1, 412a20–21). As in the case of other natural existents, the formal cause of human beings, the psyche, is also their efficient and final cause. Aristotle likened the functional properties that constitute the psyche of a human being to the capacity for sight that is the function of the eye: "Suppose that the eye were an animal—sight would have been its soul" (*De Anima*, II, 1, 412b18–19).

Aristotle claimed that plants and animals also have a psyche, ordered according to the hierarchical scale of nature. At the lowest level, there is the **nutritive psyche**, the essence of plants, which serves the functions of growth, self-maintenance through nutrition, and reproduction. The **sensitive psyche,** which is the essence of animals, is responsible for sensation, the experience of pleasure and pain, imagination and memory, and locomotion through sensuous desire. The **rational psyche** or mind (*nous*), which is the essence of human beings, serves a variety of cognitive functions, such as abstraction, deliberation, and recollection.

The functions of the rational psyche in human beings presuppose the functions of the sensitive psyche, which in turn presuppose the functions of the nutritive psyche, since the functions of each lower-level psyche serve as enabling conditions for the functions of each higher-level psyche. The self-maintenance of organisms through nutrition, for example, serves as an enabling condition for sensations received by the sensitive psyche, which in turn serve as an enabling condition for the cognitive functions of the rational psyche (since for Aristotle, all thought requires images derived from sensation). Thus, human beings have three types of psyche: a nutritive psyche by virtue of which they are self-sustaining beings, a sensitive psyche by virtue of which they are sentient beings, and a rational psyche by virtue of which they are cognitive beings.

Materialism and Psychological Explanation

Aristotle was a materialist who denied that the psyche could exist independently of the material body. Since he treated the psyche as the functional form of materially instantiated substances such as plants, animals, and humans, the question of whether the psyche could survive the destruction of the material body did not arise. It was inconceivable to him that the form of any substance could exist independently of the material it constituted *as* that particular kind of substance. For Aristotle, the rational (or sensitive or nutritive) psyche could no more exist independently of the material body it constituted as the substance of a human being than the shape of a statue of Zeus could exist independently of the marble it constituted as a statue of Zeus or than the seeing of the eye could exist apart from the material of the eye:

> As the pupil *plus* the power of sight constitutes the eye, so the soul plus the body constitutes the animal.
>
> From this it is clear that the soul is inseparable from its body.
>
> —(*De Anima*, II, 1, 413a2–4)

Although Aristotle was a materialist who held that psychological properties and capacities are instantiated in material bodies, he was not a *reductive explanatory materialist*. He did not claim that psychological states and processes could be reductively explained in terms of their material components. This comes out clearly in Aristotle's discussion of emotions such as anger, for example:

> Anger should be defined as a certain mode of movement of such and such a body (or part or faculty of a body) by this or that cause and for this or that end. . . . Hence a physicist would define an affection of soul differently from a dialectician; the latter would define e.g. anger as an appetite for returning pain for pain, or something like that, while the former would define it as a boiling of the blood or warm substance surrounding the heart. The one assigns the material conditions, the other the form or account; for what he states is the account of the fact, though for its actual existence there must be embodiment of it in a material such as it is described by the other.
>
> —(*De Anima*, I, 1, 403a26–403b4)

Although Aristotle denied that psychological functions could be reductively explained in terms of their material components and processes, he insisted that theories of psychological functions must be constrained by theories of their material instantiation (although he mistakenly believed that they are instantiated in the heart rather than the brain, whose function he believed was to cool the blood). Thus Aristotle defended the autonomy of psychological explanation while insisting that it is not divorced from the material conditions of the natural world. For Aristotle, this was a distinctive virtue of his theoretical position in contrast to dualist accounts of the immateriality of the psyche: Such accounts were not constrained in any way by the material conditions of psychological functioning. As Aristotle put it, such accounts merely "join the soul to a body, or place it in a body, without adding any specification of the reason of their union, or of the bodily conditions required for it" (*De Anima*, I, 3, 407b15–17).

Sensation, Perception, and Cognition

In his account of sensation and perception, Aristotle claimed that there are **special sensibles**, objects of sense that are only discernible by a single sense. Color is the special object of sight, sound the special object of hearing, and so on for smell, touch, and taste, the five senses recognized by Aristotle. He also claimed that there are **common sensibles**, such as movement and magnitude, which can be perceived by more than one sense. Each of the five senses requires a sense organ and a medium between the sensible quality and that organ. In the case of sight, for example, the object seen must possess color, and there must be a transparent medium containing light between the object and the eye. Aristotle maintained that error is possible with respect to judgments about common sensibles, but not with respect to judgments about the special objects of sense.

Aristotle also postulated a **common sense** that combines information about special and common sensibles to form an integrated perception of substances in the external world, such as apples and antelopes. He does not seem to have conceived of common sense as an additional sense requiring a separate organ, but as an emergent function of the five senses working in unison at a complex level of organization. Aristotle held that the most rudimentary form of sense perception is touch. He claimed that touch is linked to basic forms of desire, which are in turn linked to the capacity to experience pleasure and pain.

Aristotle believed that cognition is dependent upon imagery: "Without an image thinking is impossible" (*On Memory*, 1, 450a1). He conceived of images as representations of substances and their properties derived from our sensory experience, as faint copies or traces of sensory experience that survive in memory:

> Memory . . . is the having of an image, related as a likeness to that of which it is an image; and . . . it has been shown that it is a function of the primary faculty of sense-perception.
>
> —(*On Memory*, 1, 451a16–18)

Aristotle distinguished between simple memory, the ability to recognize an image as a representation of something in the past, and recollection, which involves the active search of memory images. He claimed that animals have the capacity for simple memory, but that only humans with a rational psyche have the capacity for recollection, which involves deliberation. According to Aristotle, deliberation is the capacity to comprehend the common form of different instances of the same type of substance by abstraction from sense experience (for example, to comprehend the common form of statues or swans by abstraction from perceived instances of them), which is the basis of all theoretical knowledge, including psychological knowledge.

Aristotle claimed that deliberative reason is a capacity unique to humans, which distinguishes them from animals and plants. Although he held that plants, animals, and humans are part of a hierarchically graded system of nature, he did not maintain that differences between humans and animals are merely differences in degree of complexity. He claimed that humans have psychological capacities such as deliberation that are not instantiated to any degree in animals, but that are different in kind from the capacities shared by humans and animals, such as sensation and desire.

In his discussion of recollection, Aristotle identified a number of principles that came to form the basis of later psychological theories. He noted that recollection is facilitated by the meaningful ordering of material to be recollected, one of the phenomena explored in Wundt's Leipzig laboratory in the late 19th century. He also noted that recollection is based upon relations of similarity, contrast, and contiguity (togetherness in space and time):

> Whenever, therefore, we are recollecting, we are experiencing one of the antecedent movements until finally we experience the one after which customarily comes that which we seek. This explains why we hunt up the series, having started in thought from the present or some other, and from something either similar, or contrary, to what we seek, or else from that which is contiguous with it.
>
> —(*On Memory*, 2, 451b17–19)

These principles, in particular the principle of contiguity (with repetition), became the mainstay of associationist psychology in Europe in the 18th and 19th centuries and of behaviorist psychology in America in the 20th century.

Like his contributions to theoretical science in general, Aristotle's contributions to psychology were wide ranging. He devoted a whole work to dreams (*On Dreams*), which he explained naturalistically in terms of the free operation of images, unconstrained by sensory inputs and rational judgment. He rejected the popular notion that dreams have religious or prophetic significance, although he acknowledged the diagnostic value of certain dreams as indicators of developing medical conditions, such as dreams of walking on fire as indicators of developing fever. He emphasized the critical role of habituation in the development of psychological traits and capacities. In his account of our aesthetic appreciation of theatrical tragedy (in the *Poetics*), he described the psychological relief produced by the cathartic expression of emotion, a notion that later played a central role in psychoanalytic theory.

In his ethical writings, Aristotle maintained that happiness derives from the proper exercise of the faculties of appetite, passion, and reason. He emphasized the virtue of mediation (the "golden mean") and held that the good life is to be attained through the subordination of appetite and passion to the control of reason. The *Nicomachean Ethics* includes subtle discussions of the psychological basis of failures of rational self-control, such as intemperance, incontinence, and weakness of will (*akrasia*). Aristotle was also the first psychological theorist to recognize the social dimensions of human psychology and behavior and famously remarked that "man is by nature a political [social] animal" (*Politics*, I, 2, 1253a2).

Active and Passive Reason

Aristotle complicated matters by maintaining that although the faculty of **passive reason** is responsible for the apprehension of universals and comprehension of first principles, it cannot achieve knowledge on its own. In a difficult set of passages in *De Anima*, he claimed that potential knowledge of universals and first principles is actualized by the operation of **active reason**. Since active reason is pure actuality, it is unchangeable and unconstrained by any temporally delimited material substance. Consequently, it can and does survive the death and destruction of those formed material substances that are human beings. As Aristotle put it, active reason is "immortal and eternal" (*De Anima*, III, 5, 430a23). Christian apologists later exploited these sorts of comments, and identified active reason with the immaterial soul that is distinct from the material body, survives death, and enjoys eternal bliss or damnation. They also identified the unmoved mover, which Aristotle postulated in the *Metaphysics* as pure form and pure actuality, with the Christian God.

Yet there is no ground for such identifications in Aristotle, and Christian notions of an immortal immaterial soul and God are quite alien to the Aristotelian worldview. Aristotle seems to have introduced both active reason and the unmoved mover to satisfy metaphysical requirements in his system, since he maintained that actuality is always prior to potentiality. He maintained that active reason, for example, enables the cognitive functions of passive reason, without which "nothing thinks" (*De Anima*, III, 5, 430a25), in much the same way as light enables colors to be seen, or "makes potential colors into actual colors" (*De Anima*, III, 5, 430a, 17).

As conceived by Aristotle, active reason and the unmoved mover are as amorphous and undifferentiated as prime matter and have none of the traditional properties that Christians ascribe to the immortal soul or God. With respect to active reason, Aristotle took pains to stress that although it survives the destruction of the material body, it retains no knowledge or memories, no pleasures or pains, and no desires, emotions, motives, or personality characteristics.

Psychology and Teleology

Although Aristotle's account of the psychology of human beings was intimately linked to his general teleological account of the natural world, it did not depend upon it. His general account of the subject matter and appropriate modes of explanation in psychology can stand alone, independently of the adequacy of his particular theories in biology and physics, which have long since been rejected on empirical grounds.

There is a very real temptation, influential in the 20th century, to suppose that the attribution of ends or purposes to human agents is no more legitimate than the attribution of ends or purposes to the motion of physical bodies or the evolution of biological species and that psychology will become properly scientific only when it abandons all explanatory references to ends and purposes (Blumberg & Wasserman, 1995). Yet many of the pioneers of the scientific revolution in the 16th and 17th centuries, such as Galileo and Newton, who rejected final causal explanations in physics in favor of efficient causal explanations, acknowledged the legitimacy of final causal explanations of human and animal behavior. Although the Darwinian revolution in biology displaced purpose from the realm of biological development in the late 19th century, many evolutionary theorists (including Darwin himself) maintained that ends and purposes play a critical role in the individual adaptation of humans and animals to their environments.

The development of modern science did involve a general displacement of final causal explanation by efficient causal explanation, but teleological explanation cannot be dismissed as inherently unscientific or irrelevant in psychology. The pioneers of information theory in the 1940s, whose work presaged the cognitive revolution in psychology of the 1950s, maintained that teleological explanation is necessary to account for the "intrinsically purposive" behavior of living organisms and machines such as torpedoes, which modify their behavior in accord with information feedback (Rosenbleuth, Wiener, & Bigelow, 1943, pp. 19–20). Contemporary cognitive psychologists recognize that a great deal of human and animal behavior is intrinsically purposive, and some maintain that this is also the case with respect to the "artificially intelligent" behavior of modern computers (Boden,1977).

Functionalism

In treating psychological states and processes as the capacities of complex material bodies, Aristotle anticipated the modern **functionalist** account of mentality (Nussbaum & Putnam, 1992; Wilkes, 1992), in which mental states are conceived of as internal states of an organism that are caused by environmental stimuli and which in turn cause other mental states and behavior. By this account, mental states such as pain, anger, and the belief that there is an object in one's path are defined as internal states with characteristic stimulus causes, mental effects, and behavioral consequences.

A distinguishing feature of the functionalist account of mental states and processes is the recognition that they can be **multiply realized** in different material systems, such as human brains or the control units of digital computers, just as the functional form of a clock can be multiply realized in sundials, water clocks, and various other mechanical devices composed of different materials. Aristotle also recognized this feature when he claimed that *different materials* could be constituted as the *same type of substance* by sharing the *same essential form:*

> In the case of things which are found to occur in specifically different materials, as a circle may exist in bronze or stone or wood, it seems plain that these, the bronze or the stone, are no part of the essence of the circle, since it is found apart from them.
>
> —(*Metaphysics*, VII, 11, 1036a30–34)

Thus a circular shape may exist in bronze, wood, or stone, and a statue of Zeus may be sculpted out of limestone, marble, or quartz.

Although Aristotle knew nothing of computers, the same principle applies to the essential forms, or computational functions, involved in cognitive operations such as addition or the memorization of serial lists. Cognitive operations such as addition may be performed on abacuses, mechanical adding machines, and digital computers, and by the human brain, composed of different materials with different modes of organization. Serial lists may be memorized by computers and by human beings, despite their differences in material composition and organization. In modern terminology, the same programs or software can be run on different forms of hardware or biological wetware.

Aristotle dismissed as "absurd" the view that the rational psyche could be instantiated in any material substance, such as that of a mouse, a tree, or a pebble. In particular, he opposed the dualist view that the psyche could be temporarily "imprisoned" in any form of physical body: "as if it were possible, as in the Pythagorean myths, that any soul could be clothed in any body" (*De Anima*, I, 3, 407b21–22). This was simply a consequence of Aristotle's claim that theoretical accounts of psychological states and processes must be constrained by our best theories of their material instantiation. For similar reasons, modern functionalists, including cognitive psychologists, deny that psychological states and processes can be realized in just any physical medium. They recognize the complexity of physical architecture required for cognitive processing.

Aristotle did maintain that we have knowledge of the rational psyche only as materially instantiated in human beings (Green, 1998). Yet he was careful to note that this contingent fact does not undermine the distinction between functional forms and their material instantiation, even though we are liable to mistakenly equate them when a functional form is as a matter of fact materially instantiated in only one known manner, as in the case of the rationale psyche:

> Of things which are *not* seen to exist apart, there is no reason why the same may not be true, e.g. even if all circles that had ever been seen were of bronze (for none the less the bronze would be no part of the form); but it is hard to effect this severance in thought. e.g. the form of man is always found in flesh and bones and parts of this kind: are these then also parts of the form and the formula? No, they are matter; but because man is not found in other matters we are unable to effect the severance.
>
> —(*Metaphysics*, VII, 11, 1036a34–1036b6)

Aristotle never considered the real possibility of full-blown rational agents other than biological human beings, since he had no knowledge of computers and did not reflect on the possibility of Martian life forms, the favored examples of modern functionalists. Yet his claim that the rational psyche is as a matter of fact instantiated only in human biological systems is entirely consistent with his functionalist position, since it is a claim maintained by a good many contemporary cognitive psychologists who recognize the limitations of computer simulations of human perception and cognition. They maintain that although rudimentary psychological capacities have been actualized in computer simulations of vision and problem solving, the only known examples of organized physical bodies capable of full-blown sentience, perception, cognitive processing, and consciousness are biological human beings.

Yet like Aristotle, such theorists still maintain the functionalist emphasis on the *autonomy* of psychological explanation. They do not feel obliged to reduce theoretical accounts of cognitive functions to theories of their neurophysiological realization, although they recognize the critical constraints imposed by such theories. In this respect they are thoroughly Aristotelian: They are materialists but not reductive explanatory materialists.

Consciousness and Vitality

Aristotle's psychology has a distinctively modern ring, insofar as he anticipated the functional form of 20th-century cognitive psychology. However, some modern commentators have complained about his neglect of the concept of consciousness:

> Concepts like that of consciousness do not figure in his conceptual scheme at all; they play no part in his analysis of perception, thought, etc. (Nor do they play any significant role in Greek thought in general.) It is this perhaps that gives his definition of the soul itself a certain inadequacy for the modern reader.
>
> —(Hamlyn, 1968, p. xiii)

Aristotle does seem to have recognized the concept of consciousness:

> He who sees perceives that he sees, and he who hears that he hears . . . so that if we perceive, we perceive that we perceive, and if we think, that we think.
>
> —(*Nicomachean Ethics,* IX, 9, 1170a29–32)

Yet it is true that he had little use for it. Still, his neglect of the concept cannot be presumed to be an inadequacy of his psychology. Although many modern theorists did come to treat consciousness as an essential feature of mentality, this is not the case with respect to contemporary cognitive psychologists, who regularly appeal to unconscious mental states and processes.

Another virtue of Aristotle's psychology was his location of theories of psychological functioning within a general biological framework. In this respect he anticipated the form of functional psychology developed by early-20th-century psychologists at the University of Chicago. James R. Angell (1869–1949), the acknowledged leader of this movement, claimed that aspects of functional psychology were "plainly discernable in the psychology of Aristotle" (Angell, 1907, p. 61). Yet in conceiving of both vital biological and cognitive psychological functions as the actuality of "bodies that have life potentially," Aristotle treated the psyche as the active principle of both life and mind. It was not until the period of the scientific revolution in Europe, when principles of mechanical (efficient causal) explanation were first extended to the "vital functions" of the "body-machine" (such as respiration and digestion), that psychological theorists first came to distinguish between the explanatory principles of life and mind.

THE ARISTOTELIAN LEGACY

The death of Aristotle marked the end of the "golden age" of Greece and Athens. His student Theophrastus (c. 371–c. 286 BCE) succeeded him as head of the Lyceum. Although a prolific writer and popular teacher, Theophrastus was overshadowed by the brilliance of his master and initiated what turned out to be a long tradition of scholarly commentary on the work of Aristotle and other Greek theorists. As the centuries progressed and Christianity arose, Aristotle's work was neglected. It was rediscovered and developed by later Islamic and medieval Christian scholars, ironically to the point that much of Aristotle's naturalistic and empirically inspired science came to be treated as religious dogma.

DISCUSSION QUESTIONS

1. Early Greek scientists and physicians frequently conceived of physical and psychological health as a form of harmony or balance, and this idea has remained popular with many later psychologists. Why do you think this is so? Why should we presume that psychological health in particular involves any form of harmony or balance?
2. Leucippus and Democritus advanced an atomic theory of nature: They maintained that all physical objects (such as planets, trees, and animals) are combinations of independent atoms. Yet unlike later scientific psychologists influenced by 18th-century atomism, they did not claim that the "elements" of perception are atomistic in nature. Democritus's theory of perception focused on the reception of form elements (such as shape) rather than perceptual atoms. Were the Greek atomists being inconsistent in this respect? Or is there no essential connection between physical and psychological atomism?
3. The Greek atomists Leucippus and Democritus conceived of a cold, hard mechanistic universe of atoms and the void, governed by rigid deterministic laws. Such a conception has proved abhorrent to many later humanistic and religious thinkers (including humanistic psychologists), who maintain that such a conception renders life meaningless and purposeless. Yet Democritus himself was not driven to despair, and maintained that the goal of human life is happiness, which he thought best achieved through self-discipline and moderation. Was he being inconsistent? Must nature be meaningful and purposive in order for us to lead meaningful and purposive lives?
4. Aristotle thought that teleological explanation applies to all physical, biological, and psychological processes, but many think that such explanations have no place in science. Some have tried to restrict teleological explanation to the explanation of biological and/or psychological processes, while others have claimed that teleological explanation applies to the behavior of some physical systems (machines) and some biological and psychological systems.

Do you think that teleological explanations have any place in contemporary science, including psychological science? If so, where do you think they apply?

5. Aristotle noted how the forms of substances can be multiply realized in different materials (a statue of Zeus may be made of marble, wood, or ice) and suggested that the same is true in principle of the functional form of the rationale psyche, even though as a matter of fact we are acquainted only with the rationale psyche of human beings. Do you think that psychological states and processes can be multiply realized—for example, in suitably developed robotic computers or alien life-forms of different material constitution? Is it just a contingent fact that the only full-blown sentient, cognitive, and conscious beings known to psychological science are human beings?

GLOSSARY

active reason In Aristotle, pure actuality that enables knowledge of universals and first principles.

animal spirits According to Alcmaeon, the material carriers of nerve impulses.

animism The belief in immaterial spirits or souls.

being vs. becoming The contrast between the view that reality is unchanging (held by Parmenides) and the view that it is constantly changing (held by Heraclitus).

common sense According to Aristotle, the faculty that combines information from the special senses into unified perception.

common sensibles According to Aristotle, properties that can be discriminated by more than one sense (e.g., movement can be discriminated by both sight and touch).

dialectic method A method of argument involving the systematic exploration of arguments for and against opposing positions.

dualism The view that the psyche (or soul or mind) and material body are distinct entities.

efficient cause The agency responsible for an existent.

eidola Faint copies of physical objects that some early Greek theorists such as Empedocles and Democritus believed emanated from physical objects and explained our perception of them.

Eleatic school The group of early Greek formalist theorists associated with Elea in southern Italy, whose members included Parmenides, Zeno of Elea, and Xenophanes.

entelechy According to Aristotle, the process by which what is merely potential becomes actual, through the realization of its form (e.g., the embryo developing into a chick).

final cause The end or function or purpose for which something exists.

formal cause The essential form of an existent.

formalism The view that the universe is best explained in terms of formal or mathematical relations.

Forms, theory of In Plato, the theory that ultimate reality is constituted by abstract ideas or Forms, in which concrete physical particulars derivatively "participate."

four causes According to Aristotle, the material, formal, efficient, and final causes of an existent.

four humors According to Hippocrates, the bodily substances that are formed from the four elements of Empedocles. Yellow bile is formed from air, blood from fire, black bile from earth, and phlegm from water. He maintained that health derives from the proper balance of these humors, and disease from their imbalance.

functionalism Theory in which mental states are conceived of as internal states of an organism that are caused by environmental stimuli and that in turn cause other mental states and behavior.

great chain of being The term used by the historian of ideas Arthur Lovejoy to describe hierarchical conceptions of nature such as Aristotle's *scala naturae*.

holistic medicine A form of medicine that emphasizes the natural healing power of the body and treats physical and psychological disorders as disorders of the whole body.

hylomorphism Aristotle's view that substances are constituted by matter (*hule*) with substantial form (*morphe*).

induction by enumeration Generalization on the basis of observed instances. For example, generalization to "All A's are B's" on the basis of observed instances of A's that are B's.

Ionian School The group of early Greek naturalistic theorists associated with the Ionian federation of city-states, whose members included Thales and Anaximenes.

material cause The material in which an existent is realized.

multiple realizability The capacity of functionally defined entities or properties to be realized in a variety of different material systems.

naturalism The view that the universe is best explained in terms of material elements and processes.

nutritive psyche According to Aristotle, the essential functional properties of plants.

passive reason According to Aristotle, the faculty responsible for the apprehension of universals and comprehension of first principles.

physis The Greek word for the fundamental element(s) (from which the term *physics* is derived).

primary qualities Qualities that physical objects have independently of our perception of them, such as size, shape, and motion.

psyche The Greek term usually translated as "soul," but without any presumed reference to an immaterial entity.

rational psyche According to Aristotle, the essential functional properties of human beings.

recollection, theory of According to Plato, knowledge is a form of remembrance of knowledge possessed by the immortal immaterial psyche, but temporarily forgotten with each cycle of reincarnation.

reductio ad absurdum argument An argument that purports to demonstrate the falsity of assumptions by demonstrating that they lead to false or absurd consequences.

relativism Theory that truth is relative to what any individual perceives or judges to be the case.

scala naturae Aristotle's hierarchical conception of nature, ranging from prime matter (pure potentiality) through increasingly complex levels of natural substances to the unmoved mover (pure actuality).

secondary qualities Qualities that are merely the effects that physical objects produce in the sense organs of sentient beings, such as color, taste, and smell.

sensitive psyche According to Aristotle, the essential functional properties of animals.

Sophists Professional teachers of rhetoric and logic in ancient Greece.

special sensibles According to Aristotle, the special objects of the individual senses, discernible by those senses alone (e.g., color is the special object of sight).

teleological science Form of science that employs explanations in terms of ends or goal states.

teleology, extrinsic Ends or purposes of a separate being (such as God).

teleology, intrinsic Ends or purposes inherent in natural processes.

temple medicine An early form of Greek medicine based upon religious beliefs and mystical practices.

universal Common property of a class of particulars (e.g., redness, the common property of red things).

unmoved mover According to Aristotle, the first cause or principle that is pure form and pure actuality, responsible for the actualization of all things.

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CHAPTER 3

Rome and the Medieval Period

WITH THE DEFEAT OF ATHENS BY SPARTA IN THE PELOPONNESIAN WAR (431–404 BCE), the Greek city-states began to disintegrate. By the time Aristotle died in 322 BCE, they had become part of the short-lived Macedonian Empire, founded by Alexander the Great. Republican Rome invaded shortly afterward, and they were eventually incorporated into the Roman Empire. The center of learning shifted from Athens to Alexandria in Egypt. In the uncertain years that followed, the confident theoretical speculations of the Greek naturalists and formalists became the object of Skepticism and Cynicism. The Hellenistic and later Roman period witnessed a turn to more practical philosophies of life, such as Epicureanism and Stoicism.

The Romans were great technologists and administrators, but contributed little to the development of science. Mystical forms of Neoplatonism became popular and informed the emergence of Christianity in the early days of the Roman Empire. When Christianity was accepted as the state religion, the works of pagan scholars such as Pythagoras and Aristotle were denigrated and condemned. With the decline of the Roman Empire, Western Europe entered what is known as the Dark Ages, a time when many of the classical Greek texts were destroyed or lost. Alexandrian scholars fled to Constantinople, then to Persia, where the classical texts were rediscovered by Islamic scholars and eventually by Christian scholars with the advent of the Crusades. During the middle and later medieval period, Aristotle's natural philosophy was integrated with Christian theology, effectively fossilizing his theories as church dogma.

Although science developed little during the medieval period, the medievals were not as hostile to it as is commonly supposed. They did not generally persecute practicing scientists and did not burn hundreds of thousands of neurotic and psychotic women whom they misdiagnosed as witches. What was distinctive about the medieval period was the general lack of interest in the empirical evaluation of scientific theories, including psychological theories. Most medieval scholars were content to develop their theories based upon classical and theological authorities.